WEEK 7 · LECTURE L3 OF 3 · UNIT III

Round 4: The Audit

Standing up analytics and auditing a hiring algorithm for bias, transparency and compliance.

Part 1 The Round 4 brief

Part 2 What an algorithmic fairness audit involves

Part 3 The four fifths test on GreenLeaf's data

Read EEOC Uniform Guidelines; NIST AI RMF (2023)

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PART 1 · THE BRIEF

What Round 4 asked of your pod

The tool

A vendor screening algorithm piloted on GreenLeaf's applicant pools.

The task

Audit it for adverse impact, transparency and compliance with the DPDP Act.

The decision

Keep, fix, add human review, or switch it off, and defend the choice.

SIGNATURE ROUND Round 4 tests whether HR can govern the tools it buys, not only purchase them.

PART 2 · THE AUDIT

Eight steps, tagged by NIST function

Step	NIST function
1 · Define the decision and the groups	Map
2 · Map every stage where the tool acts	Map
3 · Compute selection rates by group	Measure
4 · Apply the four fifths test	Measure
5 · Check error rates and proxies	Measure
6 · Decide: keep, fix, review or stop	Manage
7 · Set human review and appeal	Manage
8 · Assign ownership and re-audit dates	Govern

PART 3 · THE TEST

The four fifths test on GreenLeaf's hiring data

	Female selection rate	Male selection rate	Ratio	Result
With the algorithm (77 requisitions)	8.32%	13.88%	0.60	Fails
Without it (63 requisitions)	13.04%	13.32%	0.98	Passes

THE METHOD

Selection rate = Offers_Female ÷ Applications_Female, and the same for men. Ratio = lower rate ÷ higher rate. Below 0.80 signals adverse impact. Compare rates, never raw offer counts.

Going deeper: the theory underneath the audit

The next slides set out the NIST AI Risk Management Framework, the SMACTR internal audit, model cards, and datasheets for datasets: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · NIST (2023)

Govern, Map, Measure, Manage

THE CORE IDEA

The AI Risk Management Framework organizes AI risk work into four functions.

HOW IT WORKS

Govern sets culture and accountability; Map frames context and risks; Measure assesses them; Manage acts on them.

LIMITS AND CRITIQUE

Voluntary guidance; it says what to do, not how much is enough.

AT GREENLEAF

The audit is Measure. Without Govern (an owner) and Manage (a decision), it changes nothing.

THEORY IN DEPTH · RAJI ET AL. (2020)

SMACTR: an internal audit in five stages

THE CORE IDEA

Firms can audit their own algorithms systematically before and after release.

HOW IT WORKS

Scoping, Mapping, Artifact collection, Testing, Reflection, each producing documents.

LIMITS AND CRITIQUE

Internal auditors may lack independence from the teams they review.

AT GREENLEAF

GreenLeaf should demand the vendor's artifacts, such as training data description and test results, before renewal.

THEORY IN DEPTH · MITCHELL ET AL. (2019)

Model cards: report how a model performs, by group

THE CORE IDEA

Every model should ship with a short document on its intended use and limits.

HOW IT WORKS

Model cards report performance broken down by group and intersection, plus ethical considerations.

LIMITS AND CRITIQUE

A card is only as honest as its authors; it does not test the model.

AT GREENLEAF

Ask the ATS vendor for a model card showing error rates for women and for Tier 3 applicants.

THEORY IN DEPTH · GEBRU ET AL. (2021)

Datasheets: document where training data came from

THE CORE IDEA

Like electronic components, datasets should come with a datasheet.

HOW IT WORKS

It records why the data were collected, how, by whom, who is represented, and recommended uses.

LIMITS AND CRITIQUE

Documentation takes effort and rarely happens without a requirement.

AT GREENLEAF

If the ATS learned from past hires that were mostly male, a datasheet would have shown it.

TAKEAWAYS

Three things to carry out of this lecture

1

An audit is Measure; it needs Govern and Manage to matter.

2

Compare selection rates, never raw counts.

3

Demand documentation from vendors before you trust a model.

NEXT Dashboards and the ten-second test.